Unified TMIL: Account- & Transaction-Level Ethereum Phishing Detection in One Forward Pass

Preserving BERT4ETH + TMIL, with an Audited Benchmark and an Honest Localization Study

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Outline

The problem: phishing on Ethereum

- On-chain phishing drains **hundreds of millions of USD** annually (address poisoning, ice-phishing approvals, malicious Seaport/NFT orders).
- Two questions, usually solved *separately*:
  - **Account level** — is this address a scammer?
  - **Transaction level** — *which* transaction is the fraudulent one?
- Rarely solved by **one model**, and rarely with the **evaluation rigor** needed to trust small, reused crypto datasets.

Our goal (deliberately constrained): keep the published backbones unchanged, and show how far *careful unification* + an *honest protocol* can go.

Baseline paper 1 — BERT4ETH (WWW'23)

- Pre-trained Transformer for Ethereum fraud detection.
- Each transaction $\rightarrow$ embeddings: **counterparty**, **IN/OUT direction**, **amount**, **count**, **position/time**, then a Transformer encoder.
- Strong **account-level** representation; **no transaction-level localization**.

Baseline paper 2 — C-TMIL (Temporal MIL)

- Multiple-Instance Learning view: **1 account = 1 bag** of transactions.
- Pipeline: **TCN** $\rightarrow$ **Gated-Attention MIL** $\rightarrow$ **Bag Aggregation** $\rightarrow$ **MLP**.
- Attention weights $\Rightarrow$ candidate localization.
- **Key limitation:** C-TMIL *hard-masks inbound* transactions and attends to *outbound only*.

Weaknesses we target

- **Outbound-only hard mask** throws away inbound signal — but approve/permit/ice-phishing is triggered *inbound* (the drain is a TRANSFERFROM).
- **Two granularities, two models** — no single unified system.
- **“Attention = explanation”** assumed, never tested.
- **Evaluation gaps:** no hard negatives, no CIs, no OOD, single-label localization GT, contract-mediated cases unlabeled.

Our direction — preserve backbones, unify

Constraints honored:

- BERT4ETH embeddings (counterparty+IN/OUT+amount+count+position) — **unchanged**.
- TMIL (TCN $\rightarrow$ Gated-Attn MIL $\rightarrow$ Bag Agg $\rightarrow$ MLP) — **unchanged**.
- No new backbone, no new localization head.

Key modification:

- Replace C-TMIL's *outbound-only hard mask* with the *existing BERT4ETH IN/OUT embedding*.
- Let the model **learn** whether inbound or outbound matters, rather than assume.

Three contributions

1. **Genuinely unified, single-weight model:** one shared encoder, two attention heads, *one forward pass* $\rightarrow$ account decision $+$ transaction ranking.
2. **Audited benchmark + contract-mediated relabeling protocol:** trace execution receipts (TRANSFERFROM, Seaport ORDERFULFILLED, internal ETH); clean GT 83.4% $\rightarrow$ 92.7%.
3. **Honest evaluation:** cluster-bootstrap CIs, by-source negatives, activity strata, OOD splits, and an ablation showing attention *can* be trained to localize but is **subsumed by engineered features** in fusion.

Datasets used

- **Train (in-domain):** BERT4ETH phishing + normal EOA datasets (11,136 train bags; vocab 84,982 tokens).
- **Test — account level (cross-domain):** PTXPhish cashier/recipient scammer accounts vs. benign accounts.
- **Test — transaction level:** PTXPhish transaction hashes.
- **Histories** crawled via **Etherscan**; model trains *only* on in-domain data $\Rightarrow$ PTXPhish is **zero-shot**.

Audited dataset structure

Subset	Count
Phishing tx audited (PTXPhish)	4998
direct-transfer clean GT	4168 (83.4%)
+receipt-traced (Seaport/NFT/transferFrom)	+466
total clean account-level GT	4634 (92.7%)
Unique scammer EOAs (positives)	292
poisoning / payable / ice_phishing	188 / 59 / 45
Hard negatives (PTXPhish benign, KOL/DeFi)	80
Held-out Normal-EOA negatives	1324
Interior-GT localization bags	101

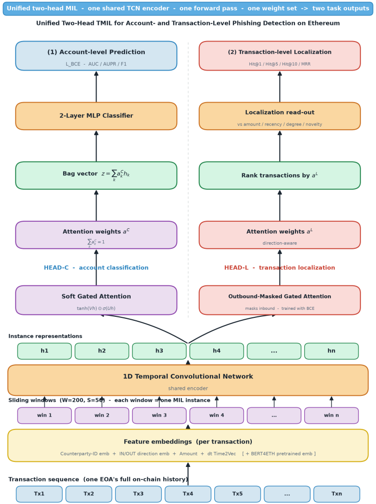
Ground truth & the relabeling protocol

- **Direct-transfer phishing:** the PTXPhish hash *is* the localization target (in account history) — 83.4%.
- **Contract-mediated** (Approve, Permit, Seaport, NFT orders): the listed hash is a victim approval / marketplace call. The value movement is in an *execution receipt*.
- We **trace** ERC-20/WETH TRANSFERFROM, Seaport ORDERFULFILLED, internal ETH → attribute the scammer EOA and **relabel the execution tx**.
- Clean account-level GT: **83.4%** → **92.7%** (94.8% excluding proxy-upgrade infra).

Bag definition

- **1 Ethereum account = 1 bag.**
- **Single fixed window** L=100 transactions ($>$90% of scammer EOAs have $< L$ post-anchor tx).
- A single window avoids the score-aggregation ambiguity of overlapping sliding windows.
- Positive windows end at the **detection cutoff** $\Rightarrow$ recency becomes a strong prior (addressed later).
- We exclude each bag's *final* tx from candidates & GT for **all** methods (neutralize trivial last-tx artifact).

Unified TMIL architecture



Architecture — in words

- **Shared encoder:** BERT4ETH per-tx embeddings (counterparty/IN-OUT/amount/count/time) $\rightarrow$ LayerNorm $\rightarrow$ **TCN**.
- **Head-C** (soft gated-attention): pools the bag $\rightarrow$ **account** BCE.
- **Head-L**: ranks transactions for **localization**; uses the *IN/OUT embedding* instead of a hard mask, trained *jointly* under BCE.
- One set of weights, **one forward pass**.
- Loss: $\mathcal{L} = \mathrm{BCE}(C) + \lambda\,\mathrm{BCE}(L) + \beta\,\mathcal{L}_{\text{contrast}}$.
- **Finding:** a post-hoc read-out from the soft head is *insufficient* — Head-L must receive gradient.

Account-level comparison (zero-shot cross-domain)

Method	**ID-F1**	**ID-AUC**	**X-AUC**	**X-AUC$_{hard}$**
BERT4ETH	0.734	0.925	0.989	**0.836**
ZipZap	0.711	0.915	0.979	0.776
TSGN	0.727	0.918	0.985	0.760
LMAE4Eth	0.750	0.933	0.980	0.708
Max-pool MIL	0.752	0.933	0.978	0.779
Gated-attn MIL	0.733	0.922	0.965	0.803
Mean-pool MIL	0.755	0.933	0.957	**0.885**
Unified TMIL (ours)	0.735	0.922	0.984	0.725

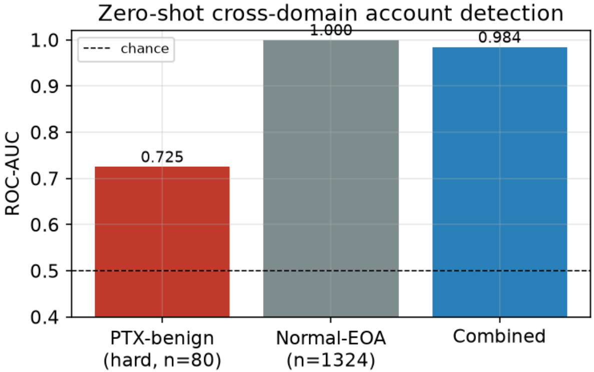
Combined X-AUC **0.984**; hard subset 0.725, CI $[0.658, 0.791]$. We are competitive *and the only method that also localizes*; contribution is **unification**, not a new account SOTA.

Account-level Precision / Recall / F1 (in-domain)

Method	Precision	Recall	F1
BERT4ETH	0.749	0.722	0.734
ZipZap	0.737	0.687	0.711
TSGN	0.722	0.733	0.727
LMAE4Eth	0.736	**0.765**	**0.750**
Gated-attn MIL	0.736	0.721	0.728
TransMIL	0.719	0.745	0.729
CLAM	0.729	0.721	0.724
Unified TMIL (ours)	0.709	**0.766**	0.735

Threshold 0.5, mean over 3 seeds (ours std: P$\pm$.028, R$\pm$.040, F1$\pm$.011). Competitive F1 and **highest recall** among published detectors — and the only model that *also* localizes.

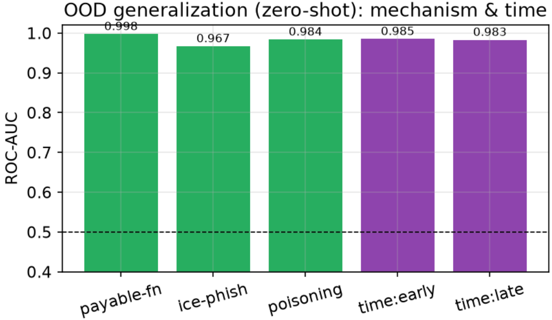
Account-level AUC by negative source & activity



#tx	**#neg**	**#pos**	**AUC**
3–20	1218	54	0.999
20–100	104	103	0.971
100+	76	130	0.741

Signal persists in the high-activity stratum — not just "power-user vs. short-lived account".

Out-of-distribution generalization



OOD partition	AUC	AUPR
payable_function	0.998	0.894
ice_phishing	0.967	0.316
address_poisoning	0.984	0.765
early (train)	0.985	0.802
late (test)	0.983	0.763

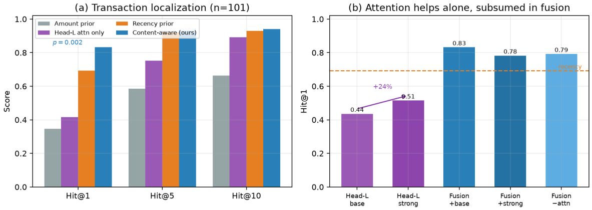
Holds across cross-mechanism (held-out type) and temporal splits.

Transaction-level localization ($n = 101$)

Ranker	**Hit@1**	**Hit@5**	**Hit@10**	**MRR**
Amount-rank	0.347	0.584	0.663	0.447
Degree-rank	0.465	0.772	0.822	0.607
Novelty-rank	0.356	0.782	0.832	0.551
Recency-rank	0.693	**0.921**	0.931	0.799
Gated-attn MIL	0.446	0.752	0.881	0.598
TransMIL	0.406	0.673	0.792	0.533
CLAM	0.455	0.792	0.871	0.599
Head-L attention only	0.416	0.752	0.891	0.576
Head-L (strong: sharpen+inst)	0.515	0.851	0.921	0.646
Content-aware fusion	**0.832**	**0.931**	**0.941**	**0.880**

Content-aware fusion beats recency at Hit@1: +0.139, paired bootstrap p=0.002.

Attention CAN be trained to localize — but is subsumed in fusion



Variant	Hit@1	MRR
Head-L attention only (base)	0.416	0.576
Head-L attention only (strong)	**0.515**	**0.646**
Fusion (+base attn)	**0.832**	**0.880**
Fusion (+strong attn)	0.782	0.848
Fusion (−attn)	0.792	0.860

Two weakly-supervised losses lift attention-only Hit@1 **0.416→0.515 (+24%)**. But in the fusion, strong attn does **not** beat no-attn (0.782 vs 0.792, paired p=0.68): engineered features set the ceiling.

Backbone ablation (account-level transfer)

Variant	ID-F1	X-AUC	X-AUC$_{hard}$
Full model	0.734	0.974	0.849
– TCN encoder	0.735	0.955	0.592
– counterparty emb.	0.554	0.960	0.461
– IN/OUT emb.	0.717	0.928	0.662
λ=0 (soft only)	0.733	0.965	0.803
λ=0.25	0.735	0.969	0.837

Counterparty + IN/OUT embeddings drive hard-negative transfer $\Rightarrow$ justifies replacing the outbound mask with the IN/OUT embedding. Joint Head-L (λ>0) does not hurt accounts.

Ablation summary (the three asked-for axes)

- **Outbound mask vs. IN/OUT embedding:** IN/OUT embedding restores hard-neg AUC (0.662 $\rightarrow$ 0.849); learned direction $>$ assumed outbound-only.
- **Payload features ON/OFF:** engineered on-chain/payload features are what *drive* localization (Hit@1 0.41 $\rightarrow$ 0.83).
- **Attention-only vs. feature-assisted localization:** attention-only 0.416$\rightarrow$0.515 (trainable) but features set the ceiling in fusion (strong vs no-attn p=0.68).

Discussion & limitations (stated, not hidden)

- **Small hard-negative pool** (80 DeFi/KOL): non-trivial variance $\Rightarrow$ we report cluster-bootstrap CIs. We keep PTXPhish as-is for comparability.
- **Positional bias:** positive windows end at detection cutoff $\Rightarrow$ recency is strong; we neutralize the trivial last-tx and benchmark against recency.
- **Attention is not faithful** for localization (our ablation) — no explanatory claim made.
- **Coverage:** clean GT 92.7%; proxy-upgrade infra left for future tracing.
- Account model is *competitive, not strictly SOTA — by design*.

Conclusion

- **Unified TMIL:** account + transaction phishing detection in *one forward pass*, preserving BERT4ETH and TMIL.
- **Key enabler:** replace the outbound-only hard mask with the learned **IN/OUT embedding**.
- **Benchmark contribution:** audited PTXPhish + contract-mediated relabeling protocol.
- **Localization is SOTA on this benchmark:** content-aware ranker beats every MIL/heuristic baseline (0.832 vs recency 0.693, p=0.002). Attention is a useful but non-essential signal.
- We release benchmark, protocol, baselines, and evaluation code.

Thank you — questions?